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Postbiotics and the gut–brain axis: A mechanistic review on modulating neuroinflammation and cognitive aging

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ABSTRACT

Aging triggers gut microbiota dysbiosis that disrupts the gut-brain axis (GBA), promoting neuroinflammation and neurodegeneration. Elderly exhibit reduced microbial diversity, depleted beneficial bacteria, and expanded pathobionts, elevating neurotoxic metabolites—lipopolysaccharides (LPS), trimethylamine-N-oxide, kynurenine derivatives, and secondary bile acids. These drive "inflammaging," blood-brain barrier breakdown, microglial activation, mitochondrial impairment, and proteinopathies in Alzheimer's and Parkinson's disease. Conversely, neuroprotective metabolites from commensals—short-chain fatty acids, indole-3-propionic acid, and urolithins—preserve gut integrity, suppress inflammation, upregulate BDNF for synaptic plasticity, and enhance mitophagy. Postbiotics, stable probiotic-derived bioactives (butyrate, polyphenol metabolites, and lactate derivatives), surpass live probiotics in safety and precision. They modulate GBA via histone deacetylase inhibition, GPR41/43 signaling, NF-κB blockade, and microglial M2 shift, blocking LPS translocation and bolstering neuronal resilience. Preclinical rodent studies demonstrate robust neuroprotection, but human translation reveals challenges: inter-individual microbiota variability (diet/genetics/comorbidities), inconsistent metabolite absorption/brain penetration between species, methodological limitations (16S rRNA vs. functional metagenomics), postbiotic standardization barriers, and sparse Phase I/II trials showing biomarker benefits without cognitive endpoints. This review synthesizes gut dysbiosis-metabolite-brain aging mechanisms, positioning postbiotics as precision therapeutics. Multi-omics stratified controlled trials are essential to validate long-term efficacy for delaying neurodegeneration and extending cognitive health.

1. Introduction

Aging is characterized by progressive physiological decline influenced by host–microbe interactions, with the gut microbiota emerging as a key modulator of both systemic and neural health (Maldonado et al., 2023). The gut–brain axis (GBA) provides metabolic, immune, and neural communication between the gut microbiome and the central nervous system (CNS) (Menezes and Shah, 2024; Alsegiani and Shah, 2022; Boehme et al., 2022). Disruption of this axis contributes to neuroinflammatory processes and is increasingly linked to age-related neurodegenerative disorders such as Alzheimer's disease (AD) and Parkinson's disease (PD) (Carabotti et al., 2015; Park and Gao, 2024). A hallmark of aging is microbial dysbiosis, marked by reduced microbial diversity, loss of beneficial commensals (e.g., Bifidobacterium, Faecalibacterium), and expansion of pathogenic taxa (e.g., Proteobacteria, Clostridia) (Khatoon et al., 2023; Mulak and Bonaz, 2015; Leite et al., 2021).

Studies show that elderly individuals exhibit decreased microbial diversity, reduced beneficial commensals, and increased pathogenic taxa (Hou et al., 2022; Ragonnaud and Biragyn, 2021; Wu et al., 2021a). These alterations lead to elevated production of pro-inflammatory and neurotoxic microbial metabolites, such as lipopolysaccharides (LPS), trimethylamine-N-oxide (TMAO), and kynurenine derivatives, which trigger neuroinflammatory pathways (Wu et al., 2021a; El Hage et al., 2023). Chronic low-grade inflammation, often referred to as "inflammaging," is a hallmark of both aging and neurodegenerative disorders,

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with gut-derived signals playing a pivotal role in this process (Saraswati and Sitaraman, 2015). Conversely, gut microbiota can also produce beneficial neuroprotective metabolites, such as short-chain fatty acids (SCFAs), polyphenol-derived metabolites, and bacterial-derived neurotransmitters (GABA, serotonin, and dopamine), which regulate neuroimmune homeostasis and synaptic plasticity (Saraswati and Sitaraman, 2015). The modulation of gut-derived metabolites is therefore a promising therapeutic target for reducing neuroinflammation and preventing cognitive decline.

In recent years, postbiotics—bioactive compounds derived from probiotic bacteria—have gained attention as potential interventions for aging and neuroprotection (Głowacka et al., 2024). Unlike probiotics, which rely on live bacterial administration, postbiotics offer greater stability, safety, and precision in modulating host physiology (Żółkiewicz et al., 2020). Emerging evidence suggests that postbiotic compounds such as butyrate, lactate, and bacterial peptides can fortify the gut barrier, suppress systemic inflammation, and enhance neuronal resilience against age-related degeneration (Głowacka et al., 2024; Scott et al., 2022).

Given the increasing global burden of neurodegenerative diseases, understanding the gut microbiota's role in brain aging and the potential of postbiotic interventions is of profound clinical and scientific significance. A metabolite-centric approach to neuroprotection may offer novel therapeutic avenues to slow down brain aging. This review aims to comprehensively explore the intricate interplay between gut microbiota, microbial metabolites, and neuroinflammation, with a particular focus on postbiotics as a next-generation strategy for healthy aging and neuroprotection.

2. The aging gut microbiome: A shift toward dysbiosis

As the body ages, the gut microbiome undergoes significant changes, leading to dysbiosis—a state of microbial imbalance characterized by a loss of diversity, an increase in harmful bacteria (pathobionts), and a decline in beneficial microbes (commensals) (Parker et al., 2022; DeGruttola et al., 2016). This shift has widespread consequences, affecting metabolism, immune function, and brain health (Parker et al., 2022). Dysbiosis contributes to chronic low-grade inflammation ("inflammaging"), a hallmark of aging that accelerates cognitive decline and neurodegeneration (Parker et al., 2022; Holmes et al., 2020; Borrego-Ruiz and Borrego, 2024). Emerging evidence suggests that gut microbial alterations actively drive aging rather than just being a consequence (Parker et al., 2022). Studies show that transplanting an aged microbiome into young mice accelerates inflammation, cognitive decline, and immune dysfunction, mimicking aging-related changes (Parker et al., 2022; Zhang et al., 2023a; Lorenzini et al., 2024). Reduced butyrate-producing bacteria and increased gut permeability further promote systemic inflammation and neurodegeneration (Bicknell et al., 2023; Chen and Vitetta, 2020; Chakraborty et al., 2024).

One of the most notable changes in the aging gut is the gradual loss of microbial diversity, which is crucial for maintaining gut integrity, immune balance, and neuroprotection (Badal et al., 2020; Loh et al., 2024). Beneficial bacteria such as *Bifidobacterium*, *Akkermansia*, and *Faecalibacterium* decline with age, reducing the production of SCFAs, particularly butyrate, which supports gut barrier function and has anti-inflammatory properties (Bradley and Haran, n.d.; Fusco et al., 2023; Xiao et al., 2024). At the same time, pro-inflammatory and opportunistic bacteria, including *Proteobacteria* and *Enterobacteriaceae*, increase, producing harmful metabolites like LPS and TMAO (Hasan et al., 2020; Askarova et al., 2020). This microbial imbalance contributes to intestinal permeability ("leaky gut"), allowing endotoxins to enter the bloodstream and reach the brain, triggering chronic neuroinflammation (Ashique et al., 2024; Kearns, 2024; Anand and Lahariya, 2025). Studies have linked gut-derived endotoxins to neurodegenerative diseases (Askarova et al., 2020). In AD, higher levels of LPS and pro-inflammatory cytokines (IL-6, TNF-α) have been found in the hippocampus, a key region for memory (Rani et al., 2022; Kinney et al., 2018; Zhao et al., 2024). PD patients show a loss of butyrate-producing bacteria, promoting α-synuclein aggregation and mitochondrial dysfunction (Zhu et al., 2022; Kalyanaraman et al., 2024). Additionally, TMAO-producing gut bacteria have been associated with vascular cognitive impairment, contributing to blood-brain barrier (BBB) disruption and cerebrovascular inflammation (Tu and Xia, 2024). This aging-associated gut dysbiosis accelerates neurodegeneration, emphasizing the GBA as a potential therapeutic target. Strategies like postbiotic supplementation, dietary interventions, and microbiome-based therapies may help restore microbial balance, reduce inflammation, and mitigate age-related cognitive decline (Boehme et al., 2022; Bradley and Haran, n.d.; Ji et al., 2023; Alam et al., 2024).

Several age-related factors accelerate gut dysbiosis, leading to chronic inflammation and neurodegeneration. Dietary changes in aging individuals, such as lower fiber and polyphenol intake, reduce beneficial microbes like *Faecalibacterium* and *Roseburia*, which produce neuroprotective SCFAs (Cronin et al., 2021; Berding et al., 2021). This depletion weakens gut barrier integrity, allowing harmful metabolites like LPS to enter circulation, triggering systemic and neuroinflammatory responses (O'Riordan et al., 2022; Wu et al., 2021b). Physical inactivity further disrupts microbial homeostasis by reducing SCFA-producing bacteria, impairing gut-brain signaling, and increasing oxidative stress (Alsegiani and Shah, 2022; O'Riordan et al., 2022). Another key factor is polypharmacy, with medications such as antibiotics, proton pump inhibitors (PPIs), non-steroidal anti-inflammatory drugs (NSAIDs), and psychotropics altering gut microbial diversity (Rogers and Aronoff, 2016; Marlicz et al., 2014). The overuse of these drugs fosters pathogenic bacteria like *Enterobacteriaceae* and *Clostridioides difficile*, while diminishing key anti-inflammatory species (Suissa et al., 2012). This imbalance exacerbates inflammaging, a chronic low-grade inflammatory state characterized by elevated IL-6, TNF-α, and IL-1β (Sanada et al., 2018). These pro-inflammatory cytokines disrupt BBB integrity, increasing susceptibility to neurotoxic metabolites and accelerating brain aging (Małkiewicz et al., 2019; Chen et al., 2023; Erdő et al., 2017). Beyond AD and PD, gut dysbiosis contributes to vascular dementia, multiple sclerosis (MS), amyotrophic lateral sclerosis (ALS), and mood disorders (Ashique et al., 2024; Ren et al., 2024; Kirby and Ochoa-Repáraz, 2018; Noor Eddin et al., 2024; Kim et al., 2022). Elevated TMAO, kynurenine metabolites, and secondary bile acids impair neuronal function, promote mitochondrial dysfunction, and drive synaptic loss, oxidative stress, and glial overactivation (Deng et al., 2024; Nagy-Grócz et al., 2024). These processes collectively fuel neurodegeneration, cognitive impairment, and psychiatric disorders, highlighting the gut microbiome as a critical factor in aging-related brain diseases (Deng et al., 2024). Restoring gut microbiota balance through dietary interventions, probiotics, and postbiotics may help counteract dysbiosis, reduce neuroinflammation, and slow brain aging, offering a potential strategy to prevent neurodegenerative diseases and promote cognitive longevity (Fig. 1).

3. Gut microbial metabolites in neurodegeneration and brain aging

The gut microbiome plays a crucial role in regulating brain health through microbial metabolites. While some of these metabolites provide neuroprotection and delay brain aging, others contribute to neuroinflammation, neurodegeneration, and cognitive decline. Dysbiosis alters the production of these metabolites, increasing the risk of neurodegenerative diseases. Understanding these mechanisms is crucial for clinicians to develop microbiome-targeted interventions for prevention and treatment of neurological disorders.

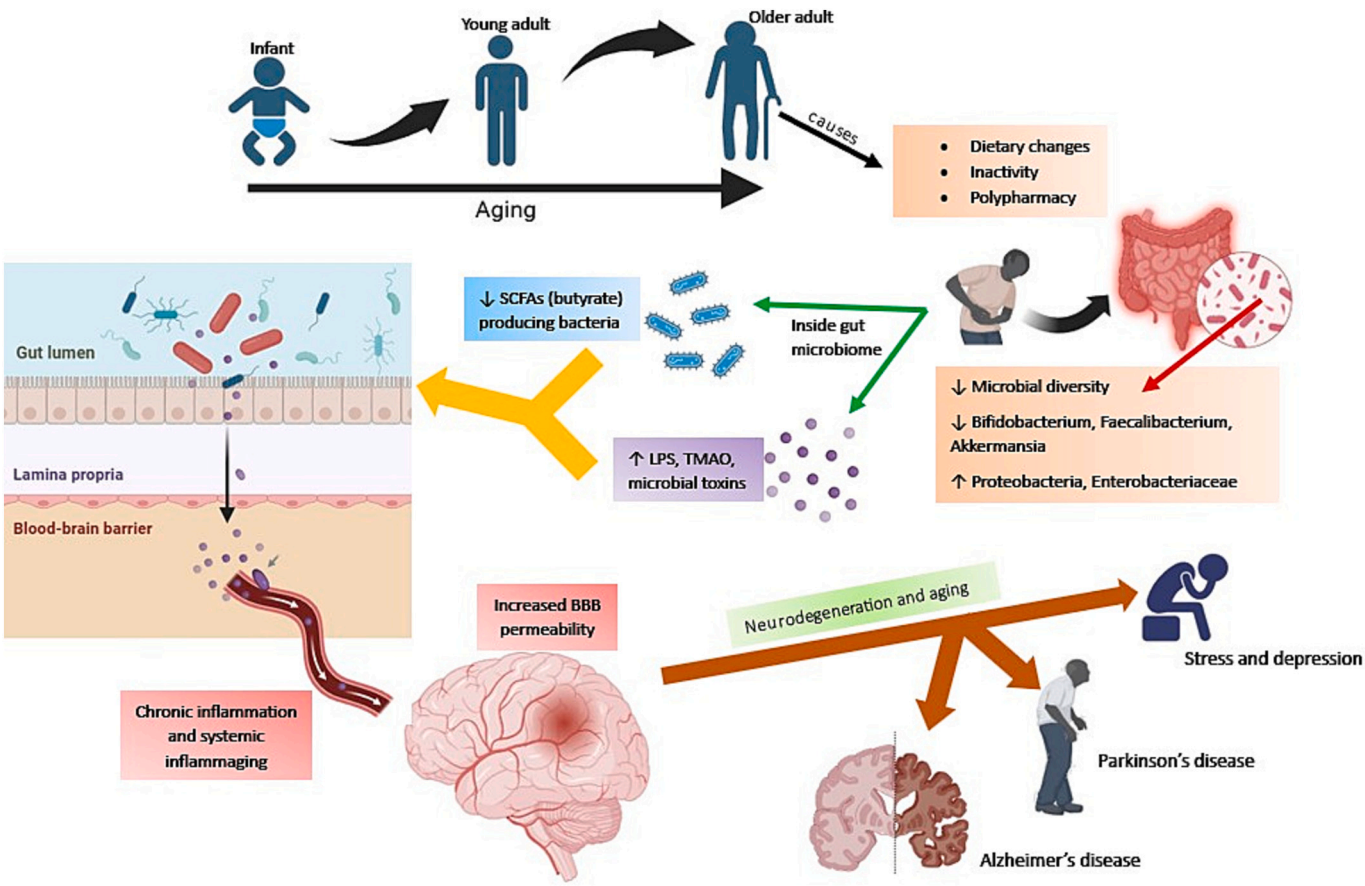

Fig. 1. Aging-driven gut dysbiosis promotes neurodegeneration. Aging alters gut microbiota composition, leading to reduced beneficial microbes, increased pathobionts, and impaired microbial metabolite production. This dysbiosis disrupts gut and blood–brain barrier integrity, promotes chronic systemic inflammation, and facilitates neuroinflammatory and neurodegenerative processes that accelerate brain aging. SCFA – Short-Chain Fatty Acids; LPS – Lipopolysaccharide; TMAO – Trimethylamine N-oxide; BBB – Blood-Brain Barrier.

3.1. Neuroprotective metabolites –

3.1.1. Short-Chain Fatty Acids (SCFAs) –

The gut microbiome plays a crucial role in health, with bacteria like *Faecalibacterium prausnitzii*, *Bifidobacterium*, *Akkermansia*, and *Lactobacillus* producing essential SCFAs—butyrate, propionate, and acetate (Fusco et al., 2023). These SCFAs are derived from dietary fiber fermentation and support gut barrier integrity, regulate immune responses, and modulate brain inflammation (O'Riordan et al., 2022; Liu et al., 2023). A balanced gut microbiome helps maintain cognitive function and prevents neuroinflammation, which is critical for healthy aging. As individuals age, microbial diversity naturally declines, leading to a reduction in SCFA production (Salazar et al., 2019). This reduction weakens gut barrier integrity, increasing intestinal permeability and triggering systemic inflammation (Fusco et al., 2023). This allows neurotoxic molecules to reach the brain, disrupting the BBB and promoting neuroinflammation. Such changes contribute to AD, PD, and MS, where chronic inflammation accelerates disease progression (Kim et al., 2025; Sweeney et al., 2018). Aging-related dysbiosis leads to a loss of beneficial microbes like *Faecalibacterium* and *Bifidobacterium*, further decreasing SCFA levels (Malik et al., 2026). The weakened gut barrier contributes to oxidative stress, mitochondrial dysfunction, and neuronal damage, key factors in cognitive decline and neurodegeneration (Kim et al., 2024; Shandilya et al., 2022; Qiao et al., 2024). Restoring microbial balance by offering a targeted approach may counteract SCFA depletion.

3.1.2. Indole derivatives –

Certain gut bacteria, including *Lactobacillus*, *Bacteroides*, and *Clostridium*, play a vital role in producing indole derivatives, particularly indole-3-propionic acid (IPA) (Hou et al., 2023; Roager and Licht, 2018). IPA is a powerful antioxidant that protects neurons from oxidative stress and inflammation (Chyan et al., 1999; Jiang et al., 2022). It helps maintain neuronal integrity and supports cognitive function by reducing reactive oxygen species (ROS) and stabilizing the BBB (Jiang et al., 2022). A well-balanced gut microbiome ensures adequate IPA production, contributing to brain health and resilience against neurodegenerative conditions. As individuals age, the gut microbiota undergoes compositional changes, leading to a decline in IPA production (Jiang et al., 2022; Randeni and Xu, 2025). Lower IPA levels have been linked to age-related cognitive decline, ALS, and depression (Randeni and Xu, 2025; Lee et al., 2024). The loss of antioxidant protection exacerbates neuronal damage, making the brain more vulnerable to degenerative processes (Randeni and Xu, 2025).

Dysbiosis often results in an overgrowth of *Proteobacteria*, which alters tryptophan metabolism (Randeni and Xu, 2025). Instead of producing beneficial indole derivatives, tryptophan is redirected toward inflammatory pathways, leading to increased production of neurotoxic metabolites (Randeni and Xu, 2025). This metabolic shift reduces neuroprotection and contributes to mood disorders and neurodegenerative diseases in elderly individuals. Modulating tryptophan metabolism through probiotic therapy is a promising approach for neuroprotection and mental health in aging populations (Gao et al., 2020). Dietary interventions rich in tryptophan, along with targeted microbiome therapies, are being explored to mitigate cognitive decline and mood

disorders in elderly individuals (Suissa et al., 2012; Zhu et al., 2024; Wang et al., 2021). These strategies hold potential for preventing neurodegeneration and improving quality of life in aging populations.

3.1.3. Urolithins –

The gut bacteria *Gordonibacter* and *Eggerthella* play a crucial role in converting dietary polyphenols, particularly ellagitannins, into urolithin A (Liu et al., 2024). Urolithin A is a potent bioactive compound that enhances mitophagy, the process of clearing damaged mitochondria (Kothe et al., n.d.). Efficient mitophagy is essential for maintaining cellular energy production and preventing the accumulation of dysfunctional mitochondria, which can drive aging and neurodegenerative diseases (Ghosh and Kumar, 2024; Banarase et al., 2023; Srivastava, 2017; Cai and Jeong, 2020). Mitochondrial dysfunction is a hallmark of aging and contributes to neurodegenerative conditions such as AD and PD (Cai and Jeong, 2020; Beal, 2005; Bustamante-Barrientos et al., 2023). In older adults, declining microbial diversity and reduced urolithin production are associated with increased neuroinflammation, synaptic dysfunction, and cognitive decline (Holmes et al., 2020).

Aging-related gut dysbiosis often leads to a decline in *Gordonibacter* levels, impairing the conversion of ellagitannins into urolithin A (He et al., 2024). This results in reduced mitochondrial quality control and increased oxidative stress within neurons, further exacerbates neuroinflammation and contributing to neurodegenerative processes (Cai and Jeong, 2020). Restoring microbial balance is key to maintaining optimal urolithin production and mitochondrial function. Urolithin A-based supplements and dietary interventions rich in ellagitannins are being explored as potential therapies for age-related neurodegeneration (Kothe et al., n.d.; D'Amico et al., 2021; Singh et al., 2022; García-Villalba et al., 2023). In elderly individuals, targeting gut microbiota composition could further optimize urolithin conversion, offering a promising strategy for brain health in aging populations (Fig. 2).

3.2. Neurotoxic metabolites –

3.2.1. Lipopolysaccharides (LPS) –

The gut microbiota, particularly *Proteobacteria* such as *Escherichia*, *Klebsiella*, and *Enterobacter*, play a significant role in systemic inflammation through the production of LPS (Zeng et al., 2017). LPS is a potent endotoxin that triggers the immune system by binding to toll-like receptor 4 (TLR4) on microglia, astrocytes, and peripheral immune cells (Ciesielska et al., 2020; McAleer and Vella, 2008). This interaction initiates a cascade of pro-inflammatory cytokine release, including IL-6, TNF-α, and IL-1β, which disrupts neuronal homeostasis (Ciesielska et al., 2020). Chronic exposure to LPS leads to sustained microglial activation, neurotoxic responses, and oxidative stress, all of which contribute to neurodegenerative diseases such as AD, PD, and vascular dementia (Lull and Block, 2010). Dysbiosis results in an overgrowth of *Proteobacteria*, increasing gut permeability, or "leaky gut" (Anand et al., 2022). This weakened gut barrier allows LPS to enter the bloodstream,

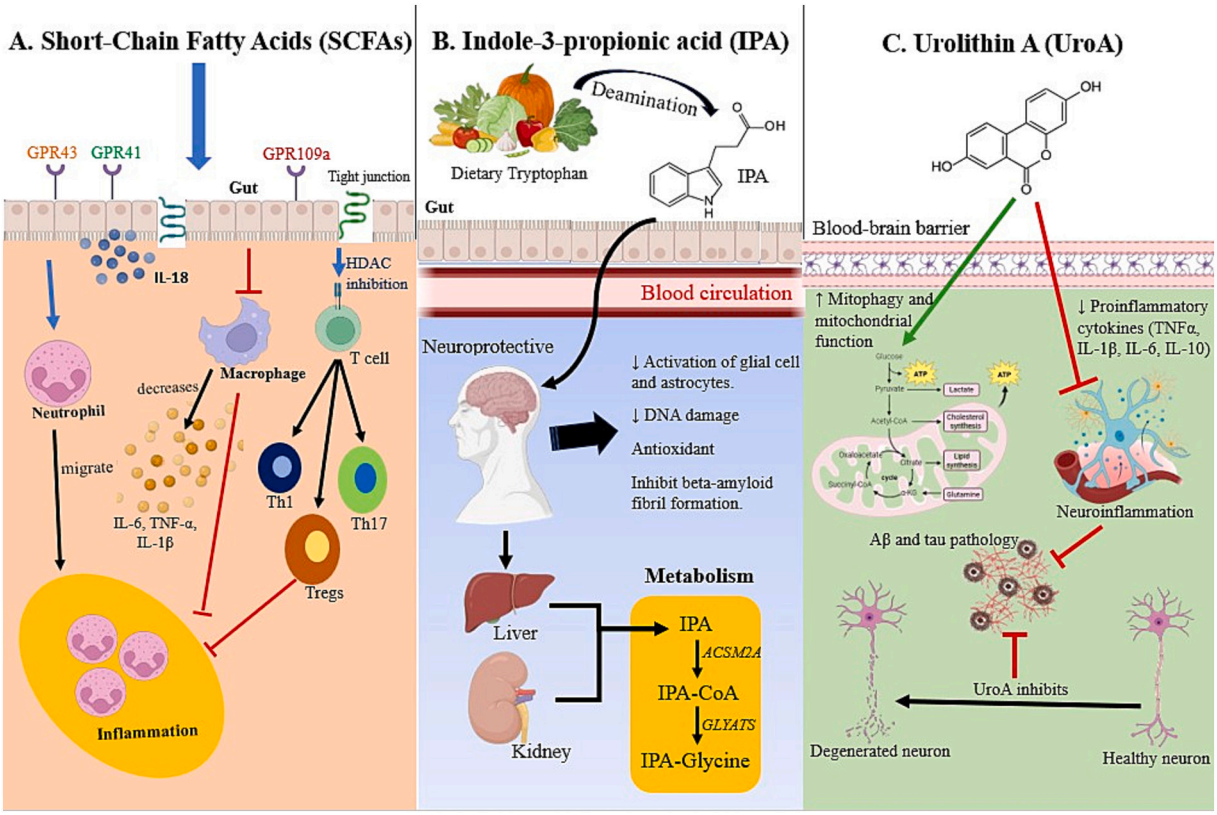

Fig. 2. Neuroprotective gut microbial metabolites and their mechanisms in brain aging. (A) Short-chain fatty acids (SCFAs), including butyrate, propionate, and acetate, are produced by commensal gut bacteria through dietary fiber fermentation and regulate gut immune homeostasis. SCFAs strengthen intestinal barrier integrity, modulate immune signaling via histone deacetylases (HDACs), Regulatory T cells (Tregs), and G-protein–coupled receptors (GPRs), suppress systemic inflammation, and preserve blood–brain barrier (BBB) function, thereby limiting age-related neuroinflammation. (B) Synthesis, circulation, and metabolism of indole-3-propionic acid (IPA) in humans. Dietary tryptophan is converted by the gut microbiota into IPA, which is absorbed across the intestinal epithelium, enters systemic circulation, and reaches peripheral organs, including the brain. IPA is subsequently metabolized and excreted via hepatic and renal pathways involving ACSM2 (acyl-CoA synthetase medium-chain family member 2 A) and GLYATS (glycine-N-acyltransferase). IPA exerts antioxidant and barrier-stabilizing effects that contribute to neuroprotection during aging. (C) Neuroprotective actions of urolithin A (UroA) in brain aging. UroA, generated by microbial conversion of dietary ellagitannins, promotes mitophagy and mitochondrial quality control, reduces neuroinflammation and oxidative stress, and has been associated with attenuation of amyloid-β and tau pathology.

triggering systemic inflammation that can reach the brain (Kalyan et al., 2022; Chmielarz et al., 2024). In individuals over 50, this process becomes more pronounced due to the decline in beneficial bacteria such as *Faecalibacterium* and *Bifidobacterium*, which otherwise help maintain gut integrity (Martín et al., 2023). Elevated circulating LPS levels have been directly linked to amyloid-beta plaque deposition in AD and α-synuclein aggregation in PD (Brown et al., 2023; Brown and Heneka, 2024). Additionally, LPS-induced systemic inflammation is associated with cerebrovascular damage, which contributes to vascular dementia and cognitive impairment (Lopez-Rodriguez et al., 2021).

Beyond inflammation, LPS exposure disrupts mitochondrial function by impairing oxidative phosphorylation and increasing the production of ROS (Casey and Murphy, 2022; Forrester et al., 2018). This mitochondrial dysfunction compromises neuronal energy metabolism, leading to synaptic degeneration, increased excitotoxicity, and eventual neuronal apoptosis (Zhao et al., 2022). Moreover, LPS influences neurotransmitter regulation by altering tryptophan metabolism, shifting it away from serotonin synthesis and toward the kynurenine pathway, which generates neurotoxic metabolites like quinolinic acid (Gao et al., 2020; Huang et al., 2023). This biochemical shift is implicated in mood disorders, cognitive dysfunction, and neuropsychiatric symptoms commonly observed in neurodegenerative diseases (Behl et al., 2021; Pathak et al., 2024). Reducing excessive LPS translocation is crucial for mitigating neuroinflammatory damage. While dietary and microbial interventions may help strengthen gut barrier integrity, the primary concern remains controlling Proteobacteria overgrowth and limiting systemic exposure to endotoxins.

3.2.2. *Trimethylamine-N-Oxide (TMAO)* –

The gut microbiota, particularly *Firmicutes* and *Proteobacteria*, including *Escherichia* and *Clostridium*, play a crucial role in the metabolism of dietary nutrients, leading to the production of TMAO (Zhou et al., 2024a). TMAO is derived from the microbial breakdown of choline, carnitine, and phosphatidylcholine, primarily found in red meat, eggs, and dairy products (Gatarek and Kaluzna-Czaplinska, 2021). Once absorbed, trimethylamine (TMA) is oxidized in the liver by flavin-containing monooxygenases (FMOs) to form TMAO (Fennema et al., 2016). Elevated TMAO levels are strongly associated with endothelial dysfunction and heightened systemic inflammation, all of which contribute to accelerated neurovascular damage (Shanmugham et al., 2023). Aging-related dysbiosis, particularly in individuals, leads to an increased abundance of *Clostridium* species, which enhances TMAO production (El Hage et al., 2023). This disrupts vascular homeostasis by promoting oxidative stress and impairing nitric oxide signaling, leading to BBB breakdown (Knox et al., 2022). The resulting endothelial dysfunction and microvascular inflammation increase the risk of vascular dementia and neurodegenerative diseases such as AD and PD (Fang et al., 2023). Additionally, TMAO has been implicated in exacerbating amyloid-beta accumulation and tau pathology, further accelerating cognitive decline (Vogt et al., 2018).

Beyond vascular impairment, TMAO influences lipid metabolism and neuroinflammation by activating pro-inflammatory pathways, including NF-κB and NLRP3 inflammasome signaling (Shanmugham et al., 2023; Zhang et al., 2021). This persistent inflammation contributes to neuronal damage and synaptic dysfunction, increasing susceptibility to neurodegenerative disorders (Adamu et al., 2024; Kempuraj et al., 2016). Furthermore, TMAO has been shown to alter mitochondrial function, leading to metabolic stress in brain cells and impairing neuronal resilience (Qiao et al., 2024). Understanding the role of TMAO in neurovascular and neurodegenerative processes highlights the importance of microbial balance in aging populations. While dietary modifications, such as reducing red meat consumption and increasing fiber intake, may help lower TMAO levels, the broader challenge lies in addressing age-related microbial shifts that predispose individuals to neurovascular complications (Aziz et al., n.d.). Identifying key microbial regulators of TMAO metabolism and their interactions with the GBA remains an essential area of research in cognitive aging and neuroprotection.

3.2.3. *Kynurenine pathway metabolites* –

The gut microbiota, particularly *Bacteroides*, *Clostridium*, and *Escherichia*, plays a central role in tryptophan metabolism, influencing neuroinflammation and cognitive function (Gao et al., 2020). Tryptophan serves as a precursor for serotonin and kynurenine pathway metabolites, which can have either neuroprotective or neurotoxic effects (Gao et al., 2020). Under homeostatic conditions, tryptophan is primarily converted into serotonin or neuroprotective kynurenic acid, both of which support mood regulation, cognitive function, and neuroprotection (Gao et al., 2020; Höglund et al., 2019). However, in a dysbiotic gut environment, microbial alterations shift tryptophan metabolism toward the production of neurotoxic compounds such as quinolinic acid (Waclawiková and El Aidy, 2018; Dehhaghi et al., 2019). Quinolinic acid is an excitotoxic metabolite that overstimulates NMDA receptors, leading to excessive calcium influx and neuronal hyperactivation (Pérez-De La Cruz et al., 2012). This cascade results in oxidative stress, mitochondrial dysfunction, and eventual neuronal apoptosis, contributing to neurodegenerative and psychiatric disorders (Pérez-De La Cruz et al., 2012). This mechanism is strongly associated with cognitive decline in Huntington's disease, schizophrenia, and depression, where excitotoxicity plays a key role in disease progression (Pathak et al., 2024).

Dysbiosis, particularly an overgrowth of *Escherichia* and *Clostridium* species, disrupts the GBA, compromising the regulation of microglial activity, leading to chronic neuroinflammatory responses that further amplify neuronal damage and cognitive impairment (Kandpal et al., 2022). The intricate relationship between gut microbiota and tryptophan metabolism underscores the need for deeper insights into microbial regulation of neuroactive metabolites. While dietary interventions and probiotics may offer potential strategies to influence tryptophan metabolism, the complexity of microbial interactions and their systemic effects on neurodegeneration require further investigation.

3.2.4. *Secondary bile acids* –

The gut microbiota, particularly *Bacteroides*, *Clostridium*, and *Bilophila*, plays a crucial role in bile acid metabolism, which directly impacts brain health (Rowland et al., 2018). Under balanced conditions, primary bile acids are synthesized in the liver and converted into secondary bile acids by gut bacteria, helping with lipid digestion and metabolic regulation (Staels and Fonseca, 2009; Winston and Theriot, 2019). However, in a dysbiotic gut environment, microbial imbalances lead to excessive production of toxic secondary bile acids, such as deoxycholic acid (DCA) and lithocholic acid (LCA), which have profound neurotoxic effects (McMillin and DeMorrow, 2016). Elevated levels of DCA and LCA contribute to neuroinflammation and mitochondrial dysfunction, two key drivers of neurodegenerative diseases (Xing et al., 2023). These bile acids disrupt neuronal membranes, impair mitochondrial energy production, and promote oxidative stress, leading to progressive neuronal damage. In AD and PD, secondary bile acid accumulation has been linked to the worsening of tau pathology, a process that accelerates cognitive decline and motor dysfunction (Ehtezazi et al., n.d.; Hurley et al., 2022).

Aging further exacerbates this imbalance, as bile acid regulation becomes less efficient in individuals over 65 years old (Majait et al., 2024). With age, shifts in gut microbiota composition—particularly an overgrowth of *Bilophila* and *Clostridium*—lead to increased production of harmful bile acids, compounding the risk of neurodegeneration (Wahlström et al., 2016; Ridlon et al., 2014). Additionally, secondary bile acids can weaken the BBB, allowing inflammatory molecules to enter the CNS, fueling chronic neuroinflammation (Quinn et al., 2014). The intricate interplay between gut microbiota, bile acid metabolism, and neurodegeneration highlights the GBA as a critical target for research. While modulating bile acid metabolism through diet and

probiotics holds potential, the complexity of these interactions necessitates further exploration (Fig. 3). Table 1 summarizes neuroprotective and neurotoxic gut microbiota metabolites, their microbial producers, effects on neurodegeneration, and underlying mechanisms.

4. Postbiotics: Emerging neuroprotective agents in gut-brain axis modulation

Postbiotics are bioactive compounds produced when probiotics ferment dietary fibers in the gut (Ma et al., 2023). Unlike probiotics, which consist of live microorganisms, postbiotics are non-viable bacterial products or metabolic byproducts that exert beneficial effects on health (Hijová, 2024). This distinction makes postbiotics more stable, safer, and easier to standardize for therapeutic applications, particularly in individuals with weakened immune systems (Wegh et al., 2019). Their neuroprotective potential is now gaining attention, as these compounds can directly or indirectly influence the GBA, modulating brain function and protecting against neurodegeneration (Aran et al., 2025; Munir et al., n.d.). One major advantage of postbiotics over probiotics is their stability. Since they do not contain live bacteria, they are not affected by storage conditions or gastric acidity (Żółkiewicz et al., 2020). This makes postbiotics a promising therapeutic alternative, particularly for individuals who may not tolerate probiotics well. Additionally, postbiotics can be tailored for specific therapeutic applications, ensuring targeted benefits (Żółkiewicz et al., 2020).

Several postbiotic compounds have been identified as key players in brain health. SCFAs, such as butyrate, propionate, and acetate, are among the most well-studied (Szydłowska and Sionek, 2022). These metabolites regulate inflammation by inhibiting histone deacetylases (HDACs), modulating the expression of genes involved in immune and inflammatory pathways (Szydłowska and Sionek, 2022). SCFAs also support gut barrier integrity, reducing the leakage of harmful bacterial endotoxins into circulation, which in turn prevents systemic and neuroinflammation (Nogal et al., n.d.; Shin et al., 2023). Polyphenol-derived postbiotics have also been recognized for their neuroprotective effects (González-Sarrías et al., 2017). These metabolites, produced from dietary polyphenols by gut bacteria, can cross the BBB and exert antioxidant, anti-inflammatory, and neurotrophic effects (Grabska-Kobyłecka et al., 2023; Vauzour, 2012). By enhancing neuronal survival pathways and mitigating oxidative stress, polyphenol-derived postbiotics hold promise in slowing cognitive decline.

Another important class of postbiotic compounds includes lactate-derived metabolites, which enhance synaptic plasticity, a crucial factor for learning and memory (Wu et al., 2023). These metabolites serve as an energy source for neurons, facilitating neurotransmission and neuronal connectivity (Wu et al., 2023). Similarly, gut bacteria also

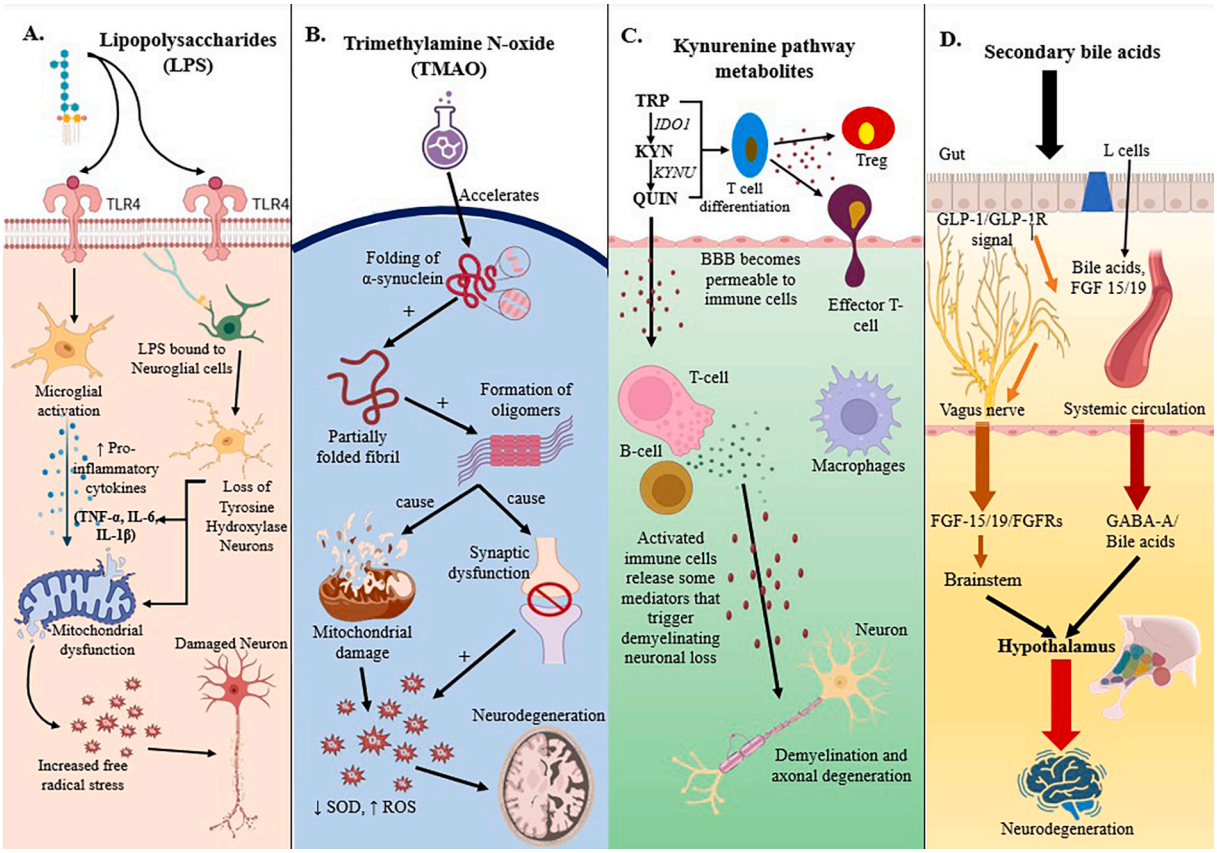

Fig. 3. Neurotoxic gut-derived metabolites and their mechanisms in neurodegeneration. (A) LPS-induced endotoxemia promotes dopaminergic neuronal death. LPS activates microglia via CD14 and directly suppresses neuronal tyrosine hydroxylase expression, leading to mitochondrial dysfunction, oxidative stress, and loss of dopaminergic neurons. (B) Mechanistic overview of neurodegeneration as trimethylamine-N-oxide (TMAO) exacerbates mitochondrial damage and free radical generation, promoting neuronal degeneration. (C) Dysregulation of the kynurenine pathway. Altered circulating kynurenine metabolites exert direct neurotoxicity on neurons and oligodendrocytes and indirectly modulate immune responses, disrupting BBB integrity, amplifying neuroinflammation, and driving demyelination, axonal loss, and neurodegeneration. (D) Gut-derived bile acids (BAs) signal to the central nervous system through direct and indirect pathways. Circulating BAs cross the BBB to activate central receptors, while BA-induced FXR–FGF15/19 and TGR5–GLP-1 signaling modulate neuroendocrine, metabolic, and neural circuits via endocrine and vagal pathways. LPS – Lipopolysaccharide; TLR4 – Toll-like receptor 4; SOD – Superoxide Dismutase; ROS – Reactive oxygen species; BBB – blood brain barrier; TRP – Tryptophan; KYN – Kynurenine; QUIN – Quinolinic Acid; KYNU – Kynureninase; IDO1 – Indoleamine 2,3-dioxygenase 1; Treg – Regulatory T cells; FXR – Farnesoid X receptor; FGF – Fibroblast growth factor; FGFR – Fibroblast growth factor receptor; TGR5 – Takeda G protein-coupled receptor 5; GLP-1 – Glucagon-like peptide-1; GLP1R – Glucagon-like peptide 1 receptor; GABA-A – Gamma-aminobutyric acid type A.

Table 1
Gut microbiota-derived metabolites and their effects on neurodegeneration.

Metabolite	Produced By	Effects	Functions
Neuroprotective			
Short-Chain Fatty Acids (Butyrate, Propionate, Acetate)	*Faecalibacterium, Bifidobacterium, Akkermansia, Lactobacillus*	Anti-inflammatory, enhances GBA	Supports BBB stability, maintains microglial homeostasis, promotes synaptic plasticity, and facilitates neuroprotection.
Indole Derivatives (Indole-3-propionic acid)	*Lactobacillus, Bacteroides, Clostridium*	Antioxidant, stabilizes neurons	Preserves neuronal structure, reduces oxidative injury, stabilizes BBB, and supports neurotransmitter balance.
Urolithins (Urolithin A)	*Gordonibacter, Eggerthella*	Supports mitochondrial health, reduces inflammation	Improves cellular energy balance by promoting mitochondrial turnover, protects against oxidative injury, and maintains neuronal function.
Neurotoxic			
Lipopolysaccharides (LPS)	*Proteobacteria (Escherichia, Klebsiella, Enterobacter)*	Triggers inflammation, disrupts neurons	Provokes systemic and neuroinflammatory responses, weakens BBB integrity, and impairs neuronal viability.
Trimethylamine-N-Oxide (TMAO)	*Firmicutes, Proteobacteria (Escherichia, Clostridium)*	Promotes neurovascular damage, accelerates cognitive decline	Compromises BBB, worsens vascular inflammation, and contributes to cognitive impairment.
Kynurenine Pathway Metabolites (Quinolinic Acid)	*Bacteroides, Clostridium, Escherichia*	Excitotoxic, contributes to neurodegeneration	Drives excitotoxic neuronal injury, promotes oxidative burden, and accelerates neurodegenerative processes.
Bile Acids (Deoxycholic Acid, Lithocholic Acid, Cholic Acid, Taurocholic Acid)	*Bacteroides, Clostridium, Lactobacillus*	Disrupts gut barrier, induces neuroinflammation	Weakens intestinal and BBB barriers, amplifies neuroinflammation, and impairs neuronal signaling and survival.

produce neurotransmitters such as GABA and serotonin, which regulate mood, cognition, and stress responses, further reinforcing the link between gut health and brain function (Miri et al., 2023; Dicks, 2022). Clinical studies have started to validate these findings. Several studies indicate that postbiotic intake improves cognition and memory, providing hope for future applications in neurodegenerative disorders. With accumulating evidence, postbiotics represent a promising frontier in neuroprotection, offering a stable, effective, and safe approach to preserving brain health and function.

5. Mechanisms of postbiotic-induced neuroprotection

Postbiotics exert their neuroprotective effects through multiple interconnected mechanisms that involve gut inflammation modulation, enhancement of gut barrier integrity, and direct effects on brain function (Ji et al., 2023). One of the primary ways postbiotics protect the brain is by reducing gut inflammation, which, if left unchecked, can contribute to systemic inflammation and neurodegeneration (Zhou et al., 2024b). Chronic low-grade inflammation originating from the gut can exacerbate neurological diseases such as AD and PD. By decreasing pro-inflammatory cytokine production and improving gut microbiota balance, postbiotics help mitigate this inflammatory burden. Another critical function of postbiotics is enhancing gut barrier integrity (Kocot et al., 2022). The gut barrier prevents harmful bacterial components such as LPS from leaking into circulation (Chelakkot et al., 2018). When LPS enters the bloodstream, it triggers systemic inflammation, which can cross the BBB and activate microglia, leading to neuroinflammation (Kalyan et al., 2022).

Microglia, the primary immune cells of the CNS, can shift between a pro-inflammatory M1 phenotype and an anti-inflammatory, repair-oriented M2 phenotype (Guo et al., 2022). Postbiotics help facilitate this transition through several coordinated pathways. SCFAs and microbial metabolites suppress M1-driving cytokines such as TNF-α, IL-1β, and IL-6, while enhancing anti-inflammatory cytokines including IL-10 and TGF-β (Yu et al., 2025). This shift is partly mediated by inhibition of NF-κB signaling and modulation of MAPK pathways. Additionally, certain postbiotic-derived metabolites influence microglial metabolic programming, promoting oxidative phosphorylation and reducing glycolysis—metabolic signatures associated with M2 polarization (Shokr, 2025). Through these actions, postbiotics help counter neuroinflammation, promote debris clearance, and support tissue repair within the CNS.

SCFAs, a major group of postbiotics, strengthen tight junctions in the gut epithelium, reducing LPS translocation and thereby lowering the risk of neuroinflammatory diseases (Pérez-Reytor et al., 2021; Ragavan and Hemalatha, 2023). SCFAs, particularly butyrate, propionate, and acetate, exert neuroprotective effects through G-protein–coupled receptor (GPR41/FFAR3 and GPR43/FFAR2) activation and epigenetic regulation (Silva et al., 2020). When SCFAs bind GPR41/43 on intestinal epithelial cells and immune cells, they initiate downstream signaling that reduces NF-κB activation, lowers pro-inflammatory cytokine production, and promotes secretion of anti-inflammatory mediators (Silva et al., 2020). This receptor-mediated mechanism contributes to strengthening the gut barrier and reducing peripheral inflammation that can otherwise propagate to the brain. In parallel, butyrate functions as a potent inhibitor of HDACs. By preventing HDAC activity, butyrate increases histone acetylation, relaxes chromatin structure, and enhances the transcription of genes involved in neurotrophic support, synaptic plasticity, and antioxidant defense (Diniz et al., 2025).

SCFAs, particularly butyrate, have been shown to increase the production of brain-derived neurotrophic factor (BDNF), a protein essential for neuronal growth, survival, and synaptic plasticity (Ragavan and Hemalatha, 2023; Church et al., 2023). Postbiotics contribute to neuroprotection by enhancing BDNF expression and improving mitochondrial homeostasis. Butyrate-driven HDAC inhibition is central to BDNF upregulation, as increased histone acetylation at BDNF promoter regions enhances transcription (Kim et al., 2009). Elevated BDNF supports synaptic plasticity, learning, memory consolidation, and neuronal survival.

Postbiotics also influence mitochondrial quality control, particularly mitophagy, the selective removal of damaged mitochondria (Prajapati et al., 2025). SCFAs activate pathways involving AMPK and PGC-1α, which enhance mitochondrial biogenesis while promoting clearance of dysfunctional mitochondria via the PINK1–Parkin axis (Prajapati et al., 2025). Improved mitophagy reduces oxidative stress, maintains ATP production, and prevents the accumulation of mitochondrial damage that contributes to neurodegenerative processes. Higher BDNF levels are associated with improved cognitive function and reduced risk of neurodegenerative diseases (Ibrahim et al., 2022). In addition, mitochondrial function in neurons is another key area where postbiotics exert their influence (Głowacka et al., 2024). Mitochondria are the energy powerhouses of cells, and their dysfunction has been implicated in various neurodegenerative disorders. By improving mitochondrial efficiency and reducing oxidative stress, postbiotics help sustain neuronal energy production and delay neurodegeneration (Fig. 4).

The role of diet and lifestyle in optimizing postbiotic benefits cannot be overstated. Consuming a fiber-rich diet enhances SCFA production, while fermented foods naturally provide beneficial postbiotic compounds (Cronin et al., 2021). The diet, rich in polyphenols, has been shown to boost gut microbiota diversity, increasing the availability of

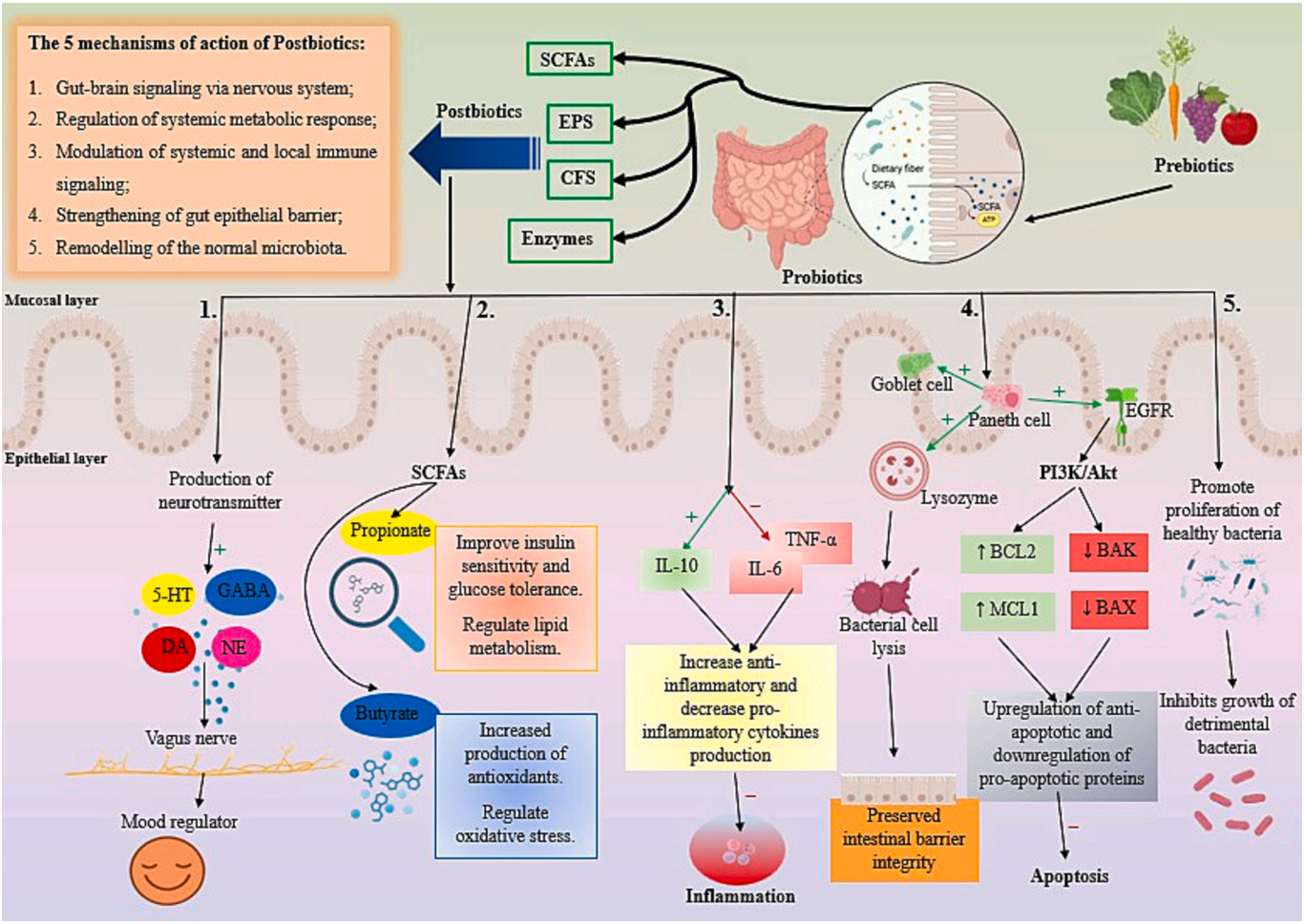

Fig. 4. Mechanisms of action of postbiotics preventing neurodegeneration. Postbiotics, including SCFAs, EPS, microbial metabolites, and CFS, exert pleiotropic effects on epithelial, immune, and neural pathways. Green arrows indicate stimulatory effects, whereas red arrows indicate inhibitory effects. 5-HT – 5-Hydroxytryptamine; BAK – BCL-2 antagonist/killer; BAX – BCL-2–associated X protein; BCL-2 – B-cell lymphoma 2; CFS – Cell-free supernatants; DA – Dopamine; EGFR – Epidermal growth factor receptor; EPS – Exopolysaccharides; GABA – Gamma-aminobutyric acid; IL – Interleukin; MCL-1 – Myeloid cell leukemia-1; NE – Norepinephrine; PI3K/AKT – Phosphatidylinositol-3-kinase/protein kinase B; SCFAs – Short-chain fatty acids; TNF-α – tumor necrosis factor-α. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

neuroprotective metabolites (Plamada and Vodnar, 2021). Exercise also plays a crucial role in shaping gut microbial composition, promoting the production of beneficial metabolites that support brain health. Additionally, maintaining a healthy sleep cycle helps regulate gut-brain interactions, ensuring optimal metabolite balance (Lin et al., 2024). Emerging research suggests that postbiotic supplementation could be a practical strategy for promoting brain health in aging individuals. Given their stability and targeted effects, postbiotics could become an integral part of future therapeutic interventions aimed at preventing neurodegeneration and supporting cognitive function in aging populations.

6. Current challenges and limitations

Despite promising preclinical evidence, several challenges limit the translational potential of postbiotic interventions for brain aging. Human microbiome responses exhibit high inter-individual variability influenced by diet, genetics, age, and comorbidities, complicating personalized dosing and efficacy prediction (Zhang et al., 2023b). Experimental outcomes also differ substantially between animal and human studies, as rodent models show consistent enhancements in BDNF expression and synaptic plasticity, whereas human trials reveal variable absorption, uncertain brain penetration, and inconsistent cognitive improvements. Most studies rely on animal models or short-term human trials, where postbiotic effects on long-term cognitive outcomes remain unproven (Zhang et al., 2023b; Bulacios et al., 2023). These issues highlight the need for delivery systems such as nanoparticle encapsulation or enteric-coated formulations to improve metabolite stability and bioavailability.

Methodological gaps further hinder progress. Microbiome analyses often prioritize 16S rRNA sequencing over functional metagenomics, failing to establish causal links between specific microbial producers, metabolite levels, and neuroinflammatory outcomes in vivo (Patricio-Martínez et al., 2025). Postbiotic standardization remains a significant barrier, as varying fermentation conditions, bacterial strains, and substrate compositions yield inconsistent bioactive profiles, undermining reproducibility across studies (Ma et al., 2023). Observational findings remain correlative; while fecal microbiota transplants from aged donors induce inflammation and cognitive decline in young recipients, reverse causation—such as brain pathology altering gut–microbiota signaling—cannot be excluded without large-scale longitudinal human cohorts incorporating multi-omics profiling (Li et al., 2025).

Clinical translation faces substantial regulatory and safety hurdles. Postbiotics lack specific FDA or EMA approval for neurodegenerative indications, with only preliminary Phase I/II data available from small trials in AD and stroke patients showing modest reductions in inflammatory markers but no definitive cognitive endpoints (Aran et al., 2025;

Rahimi et al., 2024). Potential risks include metabolite overload in frail elderly populations, such as hyperammonemia from certain bacterial lysates, or interactions with polypharmacy disrupting gut barrier function. Recent clinical trials, including those supplementing heat-killed Lactobacillus strains in stressed adults and postbiotic blends in cerebrovascular accident patients, report improvements in oxidative stress and mood but continue to demonstrate limited clinical validation due to the absence of powered RCTs with neuroimaging and cognitive batteries (Hamdi et al., 2025). Future research must prioritize large-scale, double-blind RCTs stratified by baseline dysbiosis signatures, integrating AI-driven metabolite profiling and cerebrospinal fluid biomarkers to bridge these gaps and enable precision postbiotic therapies tailored to GBA dysregulation.

7. Postbiotic therapy: an approach to neuroprotection and healthy brain aging

Recent research has highlighted the potential of postbiotics—bioactive compounds produced by probiotics—in promoting brain health and potentially slowing cognitive decline associated with aging. A study highlighted that postbiotics could influence neuroinflammation, oxidative stress, and mitochondrial dysfunction—key factors in the pathogenesis of neurodegenerative diseases (Ghosh and Kumar, 2024; Aran et al., 2025). Postbiotics protect dopaminergic neurons, reduce α-synuclein accumulation, and prevent amyloid-β plaque deposition (Bashir et al., 2024). Their neuroprotective effects include anti-inflammatory, anti-apoptotic, and antioxidant properties while promoting BDNF secretion (Bashir et al., 2024). Another study found the anxiolytic and antidepressant effects of heat-killed *Lactobacillus helveticus* strain MCC1848 on mice experiencing chronic stress-induced depression (Maehata et al., 2019). Another study exhibited the anti-anxiety effect of heat-killed *Enterococcus fecalis* (EC-12) supplement in healthy rodents (Kambe et al., 2020). Researchers conducted another study on male C57BL/6 mice to examine the effects of the postbiotic ADR-159, which contains a co-fermentate of heat-killed *Lactobacillus fermentum* and *Lactobacillus delbrueckii* (Warda et al., 2019). They observed noticeable alteration in the microbiota composition and a reduction in baseline corticosterone levels (Warda et al., 2019). Furthermore, another study acknowledged the positive effects of *Lactobacillus gasseri* CP2305 on young adults facing stressful situations, emphasizing the need for further research (Nishida et al., 2019; Lahariya et al., 2025). These findings suggest that early incorporation of postbiotics into diet and therapeutic strategies may serve as a promising approach to slowing brain aging, preserving cognitive function, and mitigating neurodegenerative decline.

8. Conclusion

Our review highlights the pivotal influence of gut-derived metabolites in neurodegeneration, emphasizing that gut dysbiosis contributes to the systemic and neuroinflammatory processes that accompany aging. Beneficial metabolites like SCFAs, indole derivatives, and urolithins exhibit neuroprotective effects by reducing inflammation, enhancing mitochondrial function, and modulating neurotransmission. Conversely, toxic metabolites such as TMAO, ammonia, quinolinic acid, and bile acids contribute to oxidative stress, neuroinflammation, and BBB disruption, accelerating neurodegeneration. This underscores the therapeutic potential of postbiotics in countering neuroinflammatory aging by early incorporating into the lifestyle. Integrating gut microbiome-based diagnostics into neurodegenerative disease management could enable early intervention and precision medicine approaches. Looking ahead, targeting the GBA through postbiotic therapies may offer a promising strategy to slow brain aging.

CRediT authorship contribution statement

Rijhul Lahariya: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Data curation, Conceptualization. **Gargee Anand:** Writing – original draft, Validation, Methodology, Data curation. **Bandana Kumari:** Writing – review & editing, Supervision, Resources, Data curation. **Ketan Priyadarshi:** Writing – review & editing, Visualization, Supervision, Methodology.

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No data was used for the research described in the article.

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